

PRELIMINARY STUDY

Introduction :

To build a system we need to understand the problem and look at what already exists; if not we might end up with something useless. This chapter gives an idea of the Hospital Management System project in this book. It talks about the problems that hospitals face.

It then looks at solutions that're already out there and their weaknesses. The proposed approach is then explained, focusing on how these weaknesses will be fixed. In the end it provides a Gantt chart of the project timeline. Explains the development phases. This way students can easily see how long each phase will take. The Hospital Management System project is broken down into phases to make it manageable. Each phase has a goal and timeframe. Understanding these phases helps in planning and execution. The chapter aims to provide a roadmap, for the project.

1. Project Overview :

This chapter presents an overview of the Hospital Management System project. The chapter starts with a discussion on the academic background that drives the project, and then it describes and explains the main theoretical concepts on which the healthcare domain is based. Afterward, the problem statement specifies in detail what the main inefficiencies and problems are that need to be solved through providing a new system. Next, the objectives of this project present what we are going to achieve in this system before describing in detail how the proposed platform will be designed and function.

1.1 Academic Framework :

We carried out our hospital management project during our final year for our bachelor's degree in computer science at the Higher Institute of Informatics and Mathematics of Monastir. Our hospital management project was launched on February 16, 2026. Officially on May 31, 2026. The project of hospital management took four months of hard work in the hospital of Sahloul. We spent 4 months in work in the hospital of Sahloul on a project of hospital management. This project that we worked on and wish to develop is a platform of hospital information. It would help in the better treatment of the patients as well as the doctors and the hospital managers.

1.2 Host Organization Profile :

Sahloul Hospital is a large public university hospital based in Sousse, Tunisia. Opened in 1991, it has become a major health care provider in the Sahel region of Tunisia, a hub of health care for the local population and a reference center for speciality practices. Sahloul Hospital manages on average thousands of patients and consultations on a daily basis ranging from simple consultations to complicated surgeries and after hour emergencies. The hospital consists of various departments and services in addition to its main mission of patient care including: cardiology, general surgery, neurology, and orthopedics to name a few. The hospital is also a university, and serves as a base for interns and residency programs in the field of health care. It is also a testing ground for many digital health tools.

Despite its very good reputation and the quality of the medical team, Sahloul Hospital is using predominantly manual methods based on paper for managing its patients and their appointments and their health records. This fact was at the origin of our project and represented a true motivation to use new digital tools that would have been useful for the patients, physicians, and staff.

The logo of Sahloul Hospital is shown in Figure 1.1.



Figure 1.1 : Logo of « Sahloul Hospital »

The following table summarizes the key information about the host organization where our project was carried out.

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Information	Details
Name	Sahloul University Hospital
Location	Route de la Ceinture, Sousse 4054, Tunisia
Built Area	Approx. 13,000 m ² on a ~10-hectare site
Number of Departments	28+ specialties
Buildings	5 main entities: Central building, South wing, North wing, MRI building, Pharmacy building
Phone	+216 73 369 411
Website	www.hopitalsahloul.tn

Table 1.1 : Host Organization Information

2. Problem Statement :

The health care system faces an increasing strain to service more people with fewer resources and less advanced systems. The public hospitals, for instance, are experiencing an escalation in the demand that must be satisfied by processes that have scarcely changed for the past few decades. Sahloul University Hospital, a prominent health care facility in the Sahel region, is no exception.

Currently, a significant number of hospital services still depend on manual procedures. For example, patients who wish to schedule an appointment must present themselves at the hospital . Medical charts are located in paper files at several departments and consequently require a time-consuming procedure to retrieve and often lead to loss of information. Prescriptions are written manually, medical tests are printed and given to the patients, and there is no easy access to medical information from the physicians' desks.

As is the case for most hospital environments, this system not only impacts patients but also doctors. Physicians have to face not only the high volume of patient flow but also routine work tasks such as finding the patient's history, rewriting the patient's medical files, and producing documents each time after a consultation. The lack of advanced digital tools only causes delays, reduces efficiency, and consequently leads to physicians devoting less time to

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clinical activities. Finally, managers also experience some difficulties related to the lack of a central system to monitor the staff, the activity, and the performance of the hospital.

The fact is that many modern technologies (e.g., voice recognition, artificial intelligence) are now mature enough to assist the workflow of the hospitals and reduce their workload; however, they still don't appear at the forefront of medical practice.

It is under these conditions that we establish our project.

3. Project Idea and Objectives :

The next final studies project is about the design and implementation of an intelligent hospital platform that will aim to improve management and organization of hospital processes through digital transformation.

The platform will be a single environment to provide a unified view for the hospital's stakeholders and ease access to and interaction with the services. It also wants to digitize processes such as appointments or medical data management and investigate the possibility of implementing smart features into the platform to help the health staff or patients. In fact, the project should help to increase productivity while decreasing the human workload by making the hospital services more user-friendly.

4. Study and Critique of Existing Solutions :

It is important that in any software project a study of already existing solutions be done; this analysis will allow us to know which solutions are already available, what their strong and weak points are, and how to improve on them in our solution. By looking at alternatives and understanding what kind of problems they address, we may be able to overcome their limits and propose a new solution that has an added value.

The objective of the current section is to analyze various already existing solutions in relation to hospital management and healthcare digitalization and then be able to extract the various limits to which our system will have to respond.

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4.1 Existing Solutions :

In examining the current landscape, we look at the Tunisian and world markets. Platforms in the health sector can be generally categorized into two main classes: patient-oriented and physician and clinic management platforms.

- Patient-oriented platforms are designed and implemented primarily for patients. These types of platforms allow patients to search for physicians and make

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appointments with them, consult their medical files, receive alerts, and sometimes communicate with health professionals.

- Physician and clinic management platforms are targeted at physicians and clinic owners. These platforms provide functionality for managing patient data, appointments, consultation files, prescriptions, clinic and hospital workflows, and activities. Some of the physicians and clinic management platforms integrate artificial intelligence capabilities to automate documentation and help with medical decision-making.

In the rest of this section, the major platforms in the above classes will be listed.

4.1.1 Patient-Oriented Platforms :

Under this heading, we have gathered two platforms that work differently within their respective regions:

- ❖ **Doctolib.fr** : A platform in France that was created in 2013, its widespread use is across the whole of Europe, as it enables booking doctors online and performing telemedicine consultations.

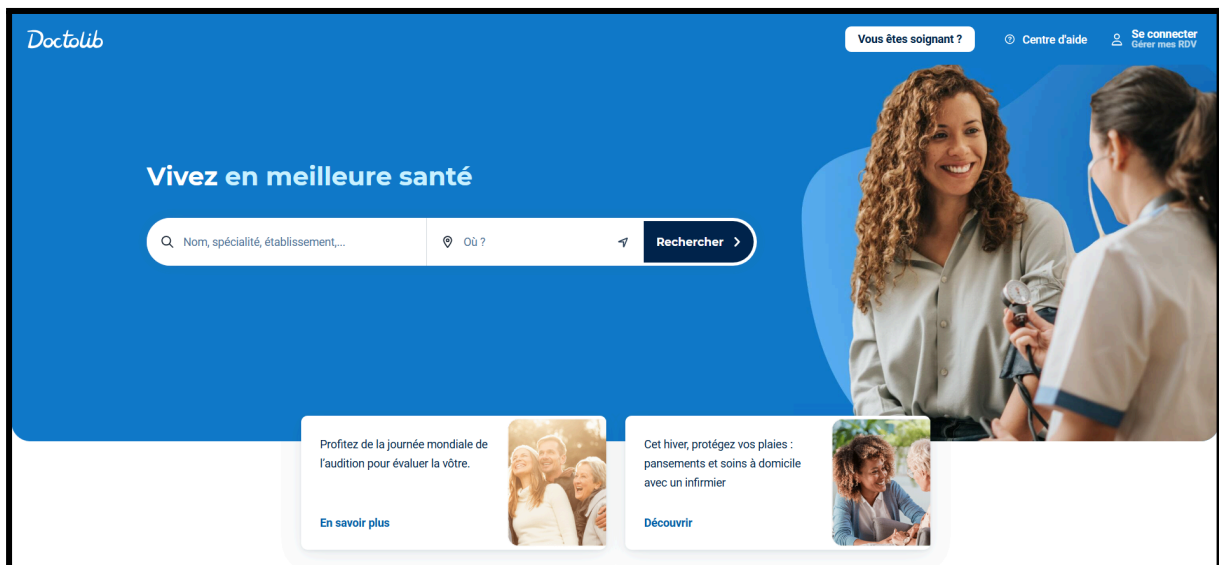


Figure 1.2 : Homepage of Doctolib.fr « www.doctolib.fr »

The platform's key features are

- Online booking of doctors : Doctors can be looked up by specialty and area, and appointments can be made 24h/24h, 7d/7d.

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- Telemedicine consultations : Doctors can see their patients by video and prescribe accordingly.
 - Appointment reminders : The platform sends notifications to avoid no-shows.
 - Doctor Dashboard : This would allow doctors to track their patients' medical records, appointments, and history of their patients.
- ❖ **Tobba.tn** : It aims to allow patients to find and book appointments with doctors in Tunisia. It allows users to get connected to the medical provider by a clean interface.

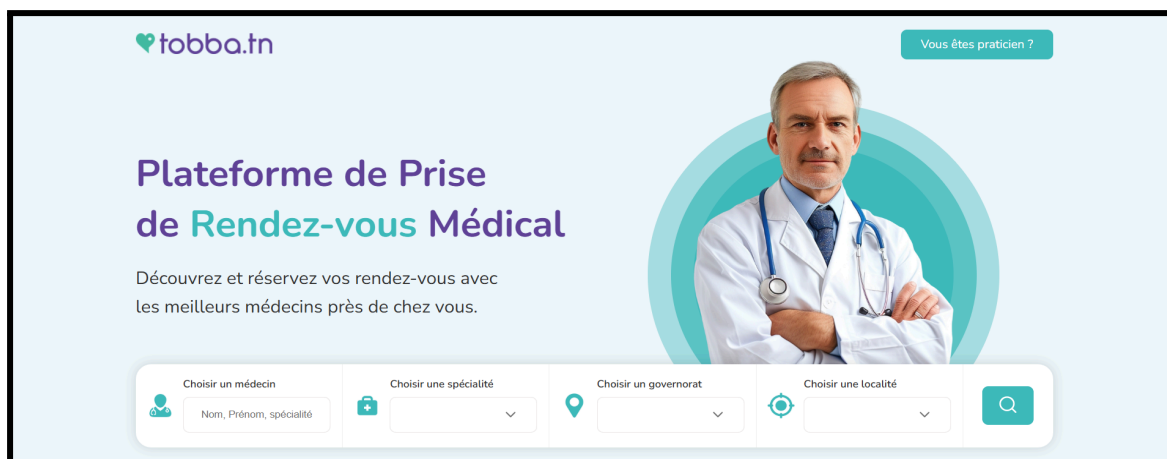


Figure 1.3 : Homepage of Tobba.tn « www.tobba.tn »

The functions available would include:

- Doctor search: Patients would be able to search for their medical specialties and browse the doctors available in their preferred location.
- Doctor profiles and availability: Patients will have access to the doctors' credentials, reviews, and doctors' availability to book the appointment.
- Online booking: Users can book appointments without making a phone call.
- Centralized directory of doctors: This feature enables doctors to have a presence on the web without the need to develop their own website, which simplifies patient accessibility and healthcare service delivery.

4.1.2 Physician and Clinical Management Platforms :

The following platforms have been selected due to their wide adoption in worldwide hospitals and health institutions:

- ❖ **Epic Systems** : Electronic health records provider used by hospitals throughout the world



Figure 1.4 : logo of Epic Systems « www.epic.com »

offers the following functionalities:

- Patient management: Management and monitoring of patient data, appointments, and billing are included in this aspect.
 - Clinical documentation: Recording consultations, medications, and patient history by the doctor through a digital document is simplified and more organized.
 - Integrated analytics: Providers can use integrated data that allows them to observe hospital performance.
 - Interoperability: Can work and connect to other hospital systems.
- ❖ **Meditech** : It is a hospital information system widely used throughout the world that includes modules such as patient registration, clinical documentation, and pharmacy operation.

The logo for Meditech, featuring the word "MEDIATECH" in a bold, green, sans-serif font, centered within a black rectangular border.

Figure 1.5 : Logo of Meditech « ehr.meditech.com »

This system has allowed the hospitals to:

- Patient registration: The software helps in registering patients in a more efficient way and improves their process of admittance
- Clinical documentation: physicians and nurses have the ability to write patient's observations, medical treatments, and consultations through a digital document.
- Laboratory and pharmacy modules: These modules integrate the processes that involve the delivery of laboratory tests and medication prescription.
- Reporting: Gives detailed administrative and clinical reports for overall monitoring and decision-making.

4.2 Evaluation of Existing Solutions :

Having analyzed in-depth the mentioned healthcare platforms, we evaluate each of them according to 8 criteria (Cx) deemed important in an adequate and complete hospital management system:

- 1. Online appointment booking (C1):** This criterion checks whether a patient can make, reschedule, or cancel medical appointments via the platform's interface and on a patient's profile..
- 2. Medical profile management (C2):** This criterion checks whether the system allows the storage and management of information relevant to the patient's medical history (illnesses, tests, allergies, etc.). The patients can, either by themselves or by a healthcare operator, view/edit this data.

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3. **Document upload and storage (C3)** : This criterion checks whether the platform provides functionality to upload and store medical reports and scans, such as results, radiography, and clinical reports, and view them.
4. **Integrated communication system (C4)** : This criterion checks if there is a platform providing communication between the patient and the physician/caregiver or among caregivers.
5. **AI-powered clinical assistant (C5)** : This criterion checks whether there are implemented features such as speech-to-text to auto-fill reports, AI analysis of test results to propose diagnoses, etc.
6. **Digital prescription generation (C6)** : This criterion checks whether doctors can generate a medical prescription in digital form, possibly with a digital signature.
7. **Indoor hospital navigation (C7)** : This criterion checks whether the platform offers patients guided indoor navigation in the hospital building.
8. **Centralized administrative dashboard (C8)** : This criterion checks whether the platform offers features enabling the administration to manage users, monitor system usage, view performance indicators, and administrate the hospital's systems.

The following table compares Doctolib, Tobba.tn, Epic Systems, and meditech against these 8 criteria.

Criteria	Doctolib.fr	Tobba.tn	Epic Systems	Meditech
C1	5/5	5/5	3/5	3/5
C2	0/5	0/5	5/5	4/5
C3	0/5	0/5	5/5	4/5
C4	3/5	0/5	4/5	3/5
C5	0/5	0/5	2/5	1/5
C6	0/5	0/5	5/5	5/5
C7	0/5	0/5	0/5	0/5
C8	2/5	0/5	5/5	5/5

Table 1.2 : Comparison of Existing Healthcare Platforms

From the comparative analysis shown in the table, it is evident that none of the current platforms cover all the desired evaluation criteria and thus validate our proposed project.

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When dealing with online appointment booking (C1), both Doctolib and Tobba.tn gain 5/5. It has become a must-have feature in every modern healthcare application, enabling patients to schedule appointments without the need to call the institution or go to the hospital. Epic Systems and Meditech obtained 3/5 for appointment scheduling. In these platforms, integration is into the institution's workflow; therefore, they are not as user-friendly or directly available to the public.

However, medical profile management (C2) entirely changes perspective, obtaining 0/5 in Doctolib and Tobba.tn. This indicates the complete absence of this feature for the patient,

meaning that there is no space for the user to keep personal information (medical data, medical history, allergies, ongoing treatments, etc.) in these platforms. Epic Systems and Meditech, however, obtain 5/5 and 4/5, respectively, as they cover perfectly the medical

profile management in their electronic health records. Yet, the latter are only available to health professionals.

The absence of document upload and storage (C3) becomes even more critical as Doctolib and Tobba.tn score 0/5. The platforms do not allow patients to upload and store their medical documents, such as prescriptions, medical reports, images, radiology reports, or clinical examinations. As a result, the users have to rely on paper forms, which can easily get lost. Epic Systems and Meditech excel again with 5/5 and 4/5, respectively. Nonetheless, this feature is usually restricted to health professionals.

Concerning integrated communication systems (C4), Doctolib is the closest to a solution with a score of 3/5; the platform is equipped with messaging services between patients and health providers, while Tobba.tn obtains 0/5. Epic systems perform even better with 4/5, as they also provide secure messaging, patient portals, and clinical communication. Meditech obtaining 3/5. The absence or inadequacy of such direct communication channels may impede patients-doctors interactions and efficiency of treatment.

It is quite surprising that all the listed platforms seem to totally ignore AI clinical assistants (C5). The only "smart" functions mentioned here are at 2/5 for Epic Systems, 1/5 for Meditech and 0/5 for both Doctolib and Tobba.tn. Despite the advances in artificial intelligence in medicine (speech recognition, report generation, diagnosis support, etc.), these services have not been yet fully integrated within the doctors' daily routine for the mentioned

applications, or they are only restricted to medical professionals for the large hospital management systems. This gap is enormous.

Likewise, the digital prescription (C6) issue seems totally neglected as it is non-existent in both Doctolib and Tobba.tn. Doctors are still using hand-written prescription forms or alternative software which leads to a lot of delays and errors. Both Epic systems and

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Meditech get a 5/5 as they offer a full e-prescribing system including digital signature and drug interaction check.

The intra-hospital navigation (C7) that could optimize the patients' time in large hospital facilities is missing in all applications from the beginning. Indeed none of the current applications allow patients to get directions through the hospital buildings to reach a particular service or a particular doctor's office.

Last but not least, concerning the centralized administrative dashboard (C8) provided for the staff, Doctolib obtain 2/5, as some basic tools are present, while Tobba.tn get 0/5. Epic Systems and Meditech, as institutional applications, have a 5/5; yet, the systems are not as user friendly and as simplistic as required in smaller institutions and less provided in Doctolib.

4.3 Proposed Solution :

Since our proposed criteria for evaluation are not satisfied by current platforms, we decided to propose a new solution with an innovation that tackles all the requirements that have been addressed. So, we propose to build and design a smart hospital platform that is a secured cloud-based hospital information management system that digitalizes the whole hospital workflow and uses artificial intelligence as a supportive feature for clinical work. The smart hospital platform provides solutions for patients, doctors, and hospital administration with the following features:

- Secured access for the medical data: We developed a robust system to manage patients' health records and data with secure authentication and authorization to protect from malicious access and ensure the confidentiality of their sensitive data.
- Flexible permission settings: Manage the access rights for the different types of users, such as a patient, doctor, or an administrator, specifying whether a user can read, write, download, upload the medical files or none of it
- Tracking activity log: A log that is updated with every user's actions on the system to achieve full control over it.
- Comprehensive reports: A report generation system that helps administrators manage their own users' activities with a transparent activity view.
- User-friendly interface: Intuitive interface for smooth navigation; patients can access all the required information about their treatments and can communicate efficiently with doctors and administrators.

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- AI-assisted clinical environment: Features such as voice detection for the medical documentation and summaries generation and suggesting diagnoses based on the patient symptoms by the integrated AI tool, which can help doctors.
- Digital prescription and drug order: Prescriptions are digitally signed by a doctor and transmitted securely; no need for handwritten documents.
- Hospital Indoor Navigation: Helps the patient to find departments or services they are looking for.
- Online appointment system: Patients can request appointments and can manage them by submitting new requests or rescheduling or canceling current appointments, which can be later approved by the doctor.

The smart hospital platform will provide advantages like:

- Better Security: For patients' data, hospital information, and all sensitive data to avoid threats and unauthorized access.
- More efficient: reduces the administrative workload by keeping medical records and appointments in a clear and straightforward manner
- Transparency: With the help of clear action logs.
- Reduce administration: Help doctors get rid of their paperwork and focus on their patients as much as possible through different AI tools.
- Good patient experience: patients feel empowered over their medical records and have easy access to their information and the hospital's service.
- Enhanced decision-making: With a transparent overview of the entire hospital's activity.

→ **Our system will have two types of applications. Mobile for the patient and web for the doctor and administrators.**



Figure 1.6 : logo of TuniSanté

5. Choice of Development Methodology :

The type of development methodology that the software is developed by has a significant bearing on the success of a software project. In this case, the Two Track Unified Process (2TUP) methodology has been applied. This is an adaptable methodology; it is iterative and promotes working between interested parties, the outcome of which is the refinement of the project throughout the process.

5.1 Unified Process :

The UP is a family of object-oriented software development process methods; they are defined by an iterative and incremental lifecycle, driven by use cases, architecture, and UML models; they emphasize risk management, user collaboration, and well-documented software architecture.

5.2 Two-Track Unified Process Methodology :

2TUP is a software development methodology based on the Unified Process. Its design is composed of three main branches:

- The functional branch: gathering of domain knowledge, users' requirements, and development of use cases.
- The technical branch: designing technical solutions, selecting techniques and architecture.
- The design branch: integrate functional and technical branches for completing the system design and implementing it.

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→ The 2TUP methodology has a Y-shaped development cycle (Figure 1.7).

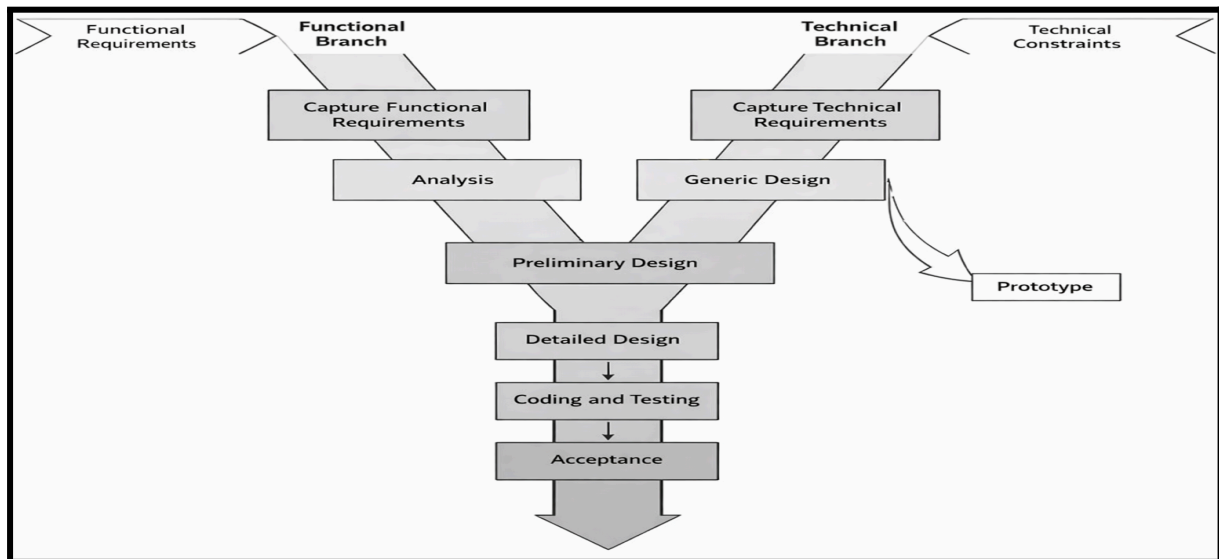


Figure 1.7 : Representation of the 2TUP Methodology

5.3 Why a Two-Track Unified Process?

For our project, we chose 2TUP because it fit the project's constraints and project requirements. This model fits well into our methodical and structured development process

from the requirement analysis to design and implementation stages. The reasons for adopting this method are as follows:

- separation of concerns: With the dual structure, the system function and technical structure are addressed at the same time.
- Iterative development: We were able to develop the platform iteratively and validate every iteration with the client and get the feedback and revise.
- Use case-driven development: It is to ensure that all the development focuses on meeting the needs of final users, which gives a concrete value.
- Flexibility and adaptation: The methodology could adapt to changes in requirements and the technical component along the project cycle, which is especially important given that healthcare technology changes rapidly.
- Project scale suitability: The iterative process was ideal for the project span and scope.

6. Gantt Diagram :

This is a project management tool that enables us to have a view of various tasks that will be achieved, how much time they are going to take, and their planning in time. A Gantt diagram is the planning of our project. It will make us be able to view the stages of the project and the duration. We have a bar the length of which represents the task time.

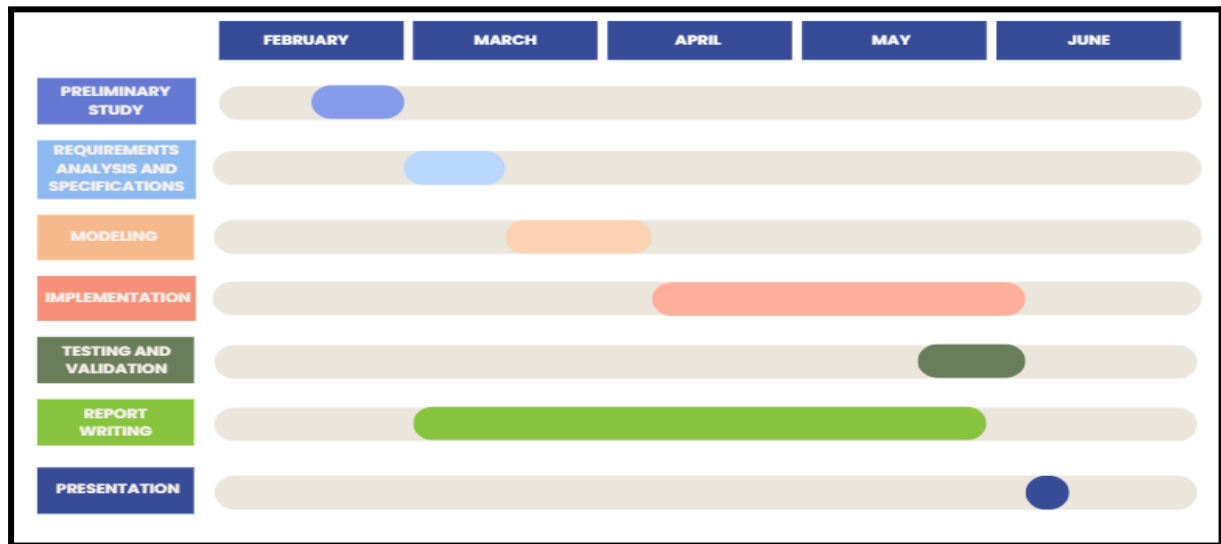


Figure 1.8 : Gantt Diagram

Conclusion :

In this chapter, we defined the general context of our end-of-studies project and the host organization. A range of existing solutions in the field of healthcare were then reviewed and discussed in light of a certain number of criteria on the basis of which we were able to clearly identify the major problems to be solved. From the results obtained, we developed a summary of the solution that is being proposed, which is TuniSante, a smart hospital platform aiming to enhance hospital administration as well as users' experience through services powered by digitalization and artificial intelligence. To end, we provided the description of the method chosen to carry out this project as well as the planning, which can be found on the Gantt chart.

Chapter 2 will concern the analysis of the specifications and system definition of the platform and detail functional and non-functional requirements.

Requirements Analysis and Specification

Introduction :

When we work on software projects we need to do things in an organized way and really understand what the user needs. and that's how we know it will be successful. The first step is to figure out what the user wants and write it down. This is called requirements analysis and specification. We have to think about everything the system can do and what it should be like. We use a way of working that helps us separate the parts of the project that are about what it does from the parts that are, about how it works. We call this the 2TUP methodology. We think this way of working will help us make a system that's very good and can be used by a lot of people. We want the system to be easy to maintain and always work well. We want the software system to be scalable. We want the software system to be maintainable.

1. General presentation:

The project of end-of-studies is about creating a hospital platform. This platform will help make hospital work easier to manage. It will also use intelligence to make things easier for doctors and patients.

The platform is for people. There is the administrator who's, in charge of the whole system. Then there is the doctor who helps patients and looks at their information. The secretary helps with appointments and medical things. The patient can also use the platform to book appointments. Look at their medical information.

We need to figure out what the platform needs to do. This is an important step. We have to make a list of what the platform should be able to do and what it should not be able to do. This is based on how all these people will use the platform. We have to think about what the administrator needs what the doctor needs, what the secretary needs and what the patient needs from the hospital platform.

2. Requirements Gathering :

The main people and things that interact with the system are being looked at. We want to make sure that the hospital runs smoothly and that people can communicate easily. So the system is being designed to make this happen. The hospital system and its features are very important. The hospital system has operations and functionalities that need to be considered. The people who use the hospital system and its features need to be able to do easily.

2.1. Patient

Patient is a primary role player, who access the system in order to make sure proper functioning of medical activities such as booking appointment, view medical history, etc.

A. Registration and Authentication: A patient can make an account with proper credentials and log in to the portal with mail ID and password, as well as recover the password.

B. Medical Profile: A patient can modify their own medical records, such as medical history, chronic diseases, allergies, and treatment.

C. Appointment Management: A patient can book/cancel/reschedule their appointment and can see their status. The secretary will check for the availability automatically.

D. Upload Medical documents: The Patient can view, delete, and upload his medical documents, such as reports, scan results, prescribe, MRI scans, etc.

E. Internal guidance system: The patient will also have access to an internal guidance system where one can find their way through the hospital and direct to the correct consulting rooms using an interactive map.

F. Emergency button: With a single tap, the patient can immediately call the emergency

2.2 Doctor

The doctor uses the platform to manage patient data, consultations, and medical decisions.

a) Patient Record Consultation : The doctor can access patient history, medical documents, and appointment details.

b) Schedule Management : The doctor can view daily schedules, monitor the waiting list in real time, and update availability.

c) AI Assistant: The system integrates artificial intelligence features to support doctors:

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- **Speech Recognition:** Converts spoken input into text.
- **Automatic Summarization:** Generates consultation reports automatically.
- **Diagnosis Suggestion:** Proposes possible diagnoses and treatments based on symptoms.

d) Prescription Generation : The doctor can generate digital prescriptions in PDF format with a digital signature.

2.3 Secretary :

The secretary plays a key role in administrative coordination and communication.

a) Appointment Management: The Secretary can schedule, cancel, or reschedule appointments on behalf of patients.

b) Patient Reception and Coordination : The secretary manages patient check-in, organizes waiting queues, and communicates with doctors.

c) General Administrative: The secretary assists in managing doctors' availability.

2.4 Administrator

The administrator is responsible for system management and overall supervision.

a) Doctor Management : The administrator can add or remove doctors, manage accounts, and handle leave schedules.

c) Monitoring and Analytics : The administrator has access to dashboards and predictive analytics.

3. Non-Functional Analysis :

The system has to be of quality and this is what the non-functional requirements are all about.

1. Security : The system is safe because it uses HTTPS for communication it encrypts passwords and it has role-based access control for people like Patients, Doctors, Secretaries who are also subAdmins and Admins. The system also logs what people do on it.

2. Performance : The platform is pretty fast. It responds in than 3 seconds it can handle hundreds of people using it at the same time and it lets people upload files quickly.

3. Availability: The platform is always available twenty four hours a day and seven days a week it makes backups of the data and it uses reliable cloud hosting.

4. Compatibility : The system has a web interface that works well on devices and it has a mobile application that is compatible with modern browsers like the ones people use every day.

5. Usability : The platform is easy to use it has an interface it is easy to navigate and it looks professional and medical.

6. Storage(neon) : The system uses PostgreSQL to store data and it uses secure storage solutions, for medical documents and it makes regular backups of the data to keep it safe.

4. Requirements Specification :

To better understand and visualize the system requirements as well as the interactions between the different actors, use case diagrams are presented in this section. These diagrams provide a clear representation of the functionalities offered by the system and illustrate how each actor interacts with the application. They also help identify the main services, user roles, and the relationships between various system components. Through these diagrams, the overall behavior of the system becomes easier to analyze, design, and implement efficiently.

4.1 Global Use Case Diagrams :

The global use case diagram illustrates the main interactions between the different actors. It highlights the functionalities available for patients, doctors, secretaries, and administrators. This diagram provides an overall view of the system's functionalities and the relationships between the different actors. Figure 2.1 illustrates the global use case diagram.

//global use-case diag//

FIG 2.1 Global Use Case Diagrams.

4.2 Detailed Use Case Diagrams :

The detailed use case diagrams show the specific actions each actor can perform in the system. They describe in a clearer way how patients, doctors, secretaries, and administrators interact with the application, helping to better understand the system's behavior and workflows.

4.2.1 Detailed Use Case Diagrams for “Mobile” :

Figure 2.2 presents the different actions that a user (patient) can perform.

//patient use-case diag//

FIG 2.2 patient Use Case Diagrams.

4.2.2 Detailed Use Case Diagrams for “Web” :

This use case diagram illustrates the interaction between three main actors. It shows how the system coordinates and supports the actions of different user roles within the same workflow, ensuring clear communication and collaboration between them.

Figure 2.2 presents the different actions that a users (Admin-Doctor-Secretary) can perform.

//web use-case diag//

5. Functional Analysis :

In this section, we identify and analyze the system requirements by describing the main use cases of the application. It includes textual explanations as well as the corresponding analysis sequence diagrams to better illustrate how the system operates.

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5.1 Textual descriptions of the main use cases :

A textual description of a use case includes its actor, preconditions, postconditions, as well as the main scenario, alternative scenarios, and exceptions. In this section, we will present the textual descriptions of the main use cases of our application.

5.1.1 Textual descriptions of “Authentication” use cases :

Table 3.1 shows the textual description of the “Authenticate” use case.

Title of use case	Authenticate
Main Actor	All Users
Description	Any user can securely authenticate to access their own space
Preconditions	The user has an account on the platform.
Postconditions	User logged in.
Main Scenario	Main scenario: Login 1. The user requests the authentication page. 2. The system displays the authentication page. 3. The user enters their email and password and clicks the “Login” button. 4. The system verifies the validity of the data entered by the user. If the email/password is incorrect, the system triggers exception 1. If a field is empty, the system triggers exception 2. 5. The system redirects the user to the home page.
Alternative Scenarios	Alternative scenario: Forgot password 1. The user clicks on the “Forgot password” button.

	2. The system asks the user to enter their email address. 3. The user enters and validates their email address. 4. The system generates a new password and sends it by email. 5. The system confirms the password reset.
Exceptions	Exception 1: If the user enters an invalid email and/or password, the system displays the error message: "Incorrect email and/or password." Exception 2: If the user leaves a field empty, the system displays the error message: "Empty field."

TAB. 3.1 - Description of the "Authenticate" use case

5.1.2 Sequence diagram of analysis for "Authenticate" :

Figure 3.1.1 illustrates the analysis sequence diagram of the "Authenticate" use case.

//Sequence diagram of analysis for "Authenticate"//

FIG. 3.1.1 – Analysis sequence diagram of "Authenticate"